The Least Squares Method Transcript

In the next video, we'll see how to use least square methods to estimate the regression line. Hello everyone, welcome to lesson 1 video 2. In this video, we'll study the least square method. This is the most common way to estimate the regression line, by minimising the total square difference between observed and predicted values.

In last video, we left off with a big question, and the question was, of all the possible lines we could draw through our data, how do we find the best one, right? How do we find the single best line, which represents the data well? So this entire session is all about finding the best fitting line. Today, we are going to answer that question that we laid down during the end of the last video, and we'll learn very clever technique that computes or that computer uses to find the perfect line, okay? First, we need a very clear way to measure how good or bad a line is. We do that by looking at the error term, okay.

Let's say for example, you have the data point here, which is the actual data point. Now we could use a line which is far away from the data and hence there would be some error and that error can be quantified as what your true value is and how is it different from what your predicted value is. Predicted value, remember, is any all of those data points on the line.

Now corresponding to this data, which is blue point here, it predicted data points is the red point here, right? Now the difference between them is called as residual or the error term. So this is what we use to define how far the actual data is from the predicted line or predicted point, right? So this is for one data point. Now what you can do is you can actually create it for all the data points and hence you will be measuring the error, which is the difference between all the actual data points and their corresponding predictions on the line.

Now we cannot use just simple differences between the true points and the predicted ones, right? So what we do is we take a square, reason being some of them could be positive, some of them could be negative and then if you add them all together, they can very well nullify, okay, they can become zero. And the other mathematical advantage that we get here is when we square error terms, you will later on see that the error square or the sum of squared of errors, this entire function, okay, it becomes convex and convex function has a very peculiar property that they have the local minima is the global minima, we'll learn about all of those things. But why we are doing sum of squares is simply because you could get some of them as negative and positives.

And if you add all of those negative and positive terms, you could get to zero, right? So in that case, you will be saying that there is a zero, but actually there would be errors, right? So that's why you take sum of squared of errors. Now the idea is to make sum of squared of errors least. That's why we call it as least squares, the method of least squares.

How do we find the best fitting line? We measure how much error each data point has with respect to its corresponding prediction. And then we try to minimise this, all of them, let's say we add the total squared error, we try to minimise this by changing the intercept or by changing the slope. See here, in this particular case, when I have the intercept, wherein the line is really far away, or let's say this one, right? The line is too far away.

The sum of squared errors is 98.25. Now the idea is to get this line aligned and the direction of the entire relationship, okay, where the association looks like. So what we do is we first try changing the intercept term, and then we might want to change the slope term. Now see how sum of squared of errors starts diminishing, right? Now this is a point wherein it doesn't change a lot.

Herein we can adjust, maybe, let's say, let's see if we can do. It seems that this actually is the best possible line, okay? So you see that the optimal value is actually 0.125. Let us see how can we go to optimal value, 1.25. Now you see that you have reached the optimal value, which is 1.25. What you have done in this entire process is to change intercept and the slope term, the beta 0 value and beta 1 value. Once you get the best approximate of beta 0 and beta 1 value, which fits the data well, when I say fits the data well, it means that the sum of squared errors, they are all 0, okay? Now this is a simple equation which we derive.

The idea is to find out the values of beta 0 and beta 1, right? So these are the equations that we calculate using calculus. We have calculus for getting to this value or we use different, different types of analytical methods to get to this value. But this is the equation that we get, beta 1 value, which is the slope term, right? So rate of, not slope term, but the coefficient attached to the slope to another variable.

And this is actually the slope term. If you think of the rate of change of Y with respect to X, this is the intercept term, right? Now if you think of beta 1 value, the equation is very much similar to covariance terms, right? Wherein you have Xi minus X bar, Yi minus pi bar and at the top and then you have how much the deviation of Xi's are from its own mean, okay? Now in this, you also get to see the correlation coefficient which is R. This Xi is the standard deviation between X and Y and this is the deviation of all X with itself, okay? So beta 1 is nothing but a ratio of a standard deviation of X and Y and the deviation of X with itself, okay? And then multiply it or scale that term with respect to the correlation coefficient. So R into XY by XX is what the equation is.

So this is what we used to find out because we generally know what X and Y are. These are the data points that we will be getting and we try to use that data to come up with the value of slope or the value of slope term. Then we go ahead and find out what is the value of the intercept term as well.

So the intercept term uses slope term, beta 0 is equals to Y bar minus beta 1 X bar. We already know what average value of Y, average of Y values that you have, what is the average of X values that you have, right? In the example that we started with in the last video, if you remember, we have university GPA and high school GPA. So this is nothing but the average of university GPA.

This is the average of high school GPA and when we take the difference of them by multiplying high school GPA with what we have got from the very first equation, okay? Then we get beta 0 which is the slope. Now in summary, we have covered or we have defined what exactly is sum of square of errors. We have also discussed the equation or how do we find the solution or how do we find the line and we define the equation of beta 1 and beta 0 and once we have both of them, we get our model which is the linear regression model and this equation could be anything.

We could get Y is equals to 0.5 plus 1.2 times X or we could get Y is equals to 100 plus 2 times X or whatever it could. It could be anything. Beta 0 and beta 1 can assume any positive negative values but then you also would have to make interpretations of what that is.

So in this case, let us say if beta 1 is negative, it means that with change in Y, with change in X, Y is decreasing or that is what that inverse relationship would mean if there is a negative sign of the slope parameter, okay? In summary, the least square methods provide the best fitting regression line by minimising squared errors. With this, we can construct a predictive equation that summarises the relationship between two variables. Next, we will learn how to assess and quantify these types of models.

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